

ZAIDH RIZME, MECHATRONICS & ROBOTICS ENGINEER

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SUMMARY

Mechatronics Engineer with hands-on experience in PCB design, 3D CAD modeling, embedded systems, computer vision, and modern programming languages including Python and C/C++. Skilled in developing electronic schematics, SolidWorks mechanical assemblies, and machine learning fundamentals. Proficient in integrating sensors, microcontrollers, ROS, MATLAB, rapid prototyping, rigorous testing, and comprehensive technical documentation.

PROFESSIONAL EXPERIENCE

Sep 2026 - Present Robotics Engineer (Intern) | Hype Invention (Pvt) Ltd.

- Developed and deployed custom automation systems and autonomous robotic platforms.
- Integrated advanced sensor pipelines for real-time navigation and control.
- Ensured high system reliability through systematic testing and diagnostics.
- Collaborated across hardware and software teams to deliver production-grade solutions.
- Optimized robotic algorithms and control loops to enhance operational speed and performance.

Jul 2025 - Feb 2026 Automation Engineer (Intern) | MAS Capital (Pvt) Ltd.

- Implemented automation systems to reduce downtime and defects in manufacturing lines.
- Integrated sensors, PLCs, and vision systems for process monitoring and quality control.
- Designed 3D CAD models and mechanical assemblies for automation systems, translating functional requirements into practical component designs.
- Assisted in testing, troubleshooting, and documentation of automation solutions.

Jul 2024 - Jul 2025 Academic Assistant | LivePhysics (Pvt) Ltd.

- Preparing and documentation of GCE Advanced Level Physics Papers, Marking Schemes, Lesson presentations.
- Coordinate with staff for scheduling and material planning.

Jan 2023 - Sep 2024 Course Instructor | NEOGRAENIUM

- Developed and implemented dynamic curriculum for LEGO robotics and Micro:Bit device.
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EDUCATION

Sep 2023 - Jul 2026 Bachelor of Engineering (Hons) in Mechatronics Engineering

University of Wolverhampton | CINEC Campus Pvt LTD

- Grade : -----

Sep 2020 - Dec 2022 GCE Advanced Level (Physical Science Stream)

Isipathana College, Colombo 05

- Combined Mathematics - B | Chemistry - B | Physics - C

PROJECTS

Dec 2025 - Jun 2026 ToF Based SLAM and Autonomous Mobile Robot | Final Year Project

- Researched ToF technology viability as a low-cost alternative to LiDAR for indoor SLAM mapping.
- Implemented sensor fusion (IMU, encoders) for pose estimation.
- Developed real-time navigation and motion control on an ESP32.
- Designed a custom 2-layer PCB for compact integration and clean power/signal routing.
- Prototyped using ROS, Gazebo transitioning to MATLAB for mapping, visualization, and analysis.

Jul 2025 - Dec 2025 Automated Modified Iron | MAS Capital Intern Project

- Developed an industrial iron with relay, timer, and temperature control for precise heating.
- Integrated pneumatic cylinder and movable arms to automate garment pre-heating before pressing.
- Improved efficiency, consistency, and reduced manual effort in branded apparel manufacturing.
- Conducted testing, troubleshooting, and optimization to ensure reliable production use.

Oct 2024 - Apr 2025 Automated Oil Level Monitoring System for Industrial Machinery

- Designed the 2-layer PCB for an ATmega328P/MS5837 oil-level monitoring system.
- Developed C firmware to process data, trigger alerts, and ensure reliable industrial performance.

Jan 2024 - Apr 2024 Wearable Active Posture Corrector Device (Research)

- Investigated the use of neodymium magnets and Hall effect sensors to detect upper body posture deviations with high sensitivity.

CORE COMPETENCIES

Hardware & Embedded	PCB Design, 3D CAD Modeling (SolidWorks), Sensor Integration, Prototyping, Hardware Testing, Machinery
Robotics & Automation	MATLAB, ROS, SLAM, Gazebo, Motion Control, Sensor Fusion
Software & Programming	Python, C/C++, Machine Learning (Fundamentals), HTML, CSS, PowerBI, Git

CERTIFICATIONS

Machine Learning *Data Manipulation, Numerical Operations, Skicit-learn, Deep Learning & AI*
Simplilearn (2026)

Git Essentials *Version control, branching strategies, repository management*
Linkedin Learning (2026)

Power BI for Beginners *Data visualization, interactive dashboards, data modeling, Power Query.*
Linkedin Learning (2026)

Python Full Course *Object-oriented programming (OOP), data structures, algorithm implementation.*
freeCodeCamp.org (2026)

SolidWorks *3D modeling, mechanical assembly design, sheet metal design, engineering drawings.*
Coursera (2026)

PROFESSIONAL AFFILIATIONS

Student Affiliate Member, Institution of Mechanical Engineers (IMechE) | ID: 80826357 | (2026)

REFEREES

Miran Dabare, MEng	Position:	Senior Executive - Autonomation Engineer
	Organization:	MAS Capital Pvt Ltd - Intimates Division
	Email:	mirand@masholdings.com
	Contact Number:	+94 77 843 7166
Dr. Sanjeewa Dharmawardana, PhD	Position:	Educational Consulting Physics Educator
	Organization:	Founder, LivePhysics (Pvt) Ltd.
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